

COMP5339 Project Assignment 2

Group Number: 80

Electricity Sector Data Streaming and Analysis

1. System Description

This project implements a data streaming system for displaying continuous power generation and emissions data from the Australian National Electricity Market (NEM). The system is designed to retrieve time-series electricity data from OpenElectricity, process it into facility-level records, publish each record as an individual MQTT message, subscribe to the live data stream, and visualise the incoming information through an interactive dashboard.

The system has several main components. First, the OpenElectricity API client is used as the data source. It retrieves facility metadata, including facility code, facility name, network region, unit code, latitude and longitude, as well as time-series measurements for power generation, CO2 emissions, regional electricity price and regional demand. The selected network is the NEM, and the data is collected at a five-minute interval.

Second, the data processing component is to clean and combine the data. Facility metadata is merged with power and emissions records through `unit_code`. Since some facilities contain multiple generation units, unit-level records are aggregated into facility-level records. This allows each facility to be shown as one record at each timestamp.

Third, the MQTT publisher converts each processed row into a JSON payload and sends it to the HiveMQ public broker through port 1883. Publishing each row separately allows the dataset to be consumed as a continuous stream rather than a static file.

Fourth, the MQTT subscription component receives the published messages from the same topic. The subscriber decodes each incoming JSON message and stores the latest records in a shared live dataset. To support real-time updates, the code uses a lock mechanism to safely manage incoming messages while the dashboard is running. The latest information is stored by facility and fuel technology, so the dashboard can keep updating the current power and emissions values for each station.

Fifth, the dashboard visualisation component is built using Dash and Dash Leaflet. It displays a map-based dashboard for the NEM regions, including New South Wales, Queensland, Victoria, South Australia and Tasmania. The dashboard adds facility markers to the map based on the received MQTT messages. Users can filter the displayed facilities by network region and fuel technology. Each marker provides key information such as station name, facility code, total power output, emissions, regional electricity price and regional demand.

Finally, the continuous execution component allows the system to simulate an unbounded data stream. The retrieval, processing and publishing process runs repeatedly with a delay between each new round of API data retrieval. The dashboard also updates regularly as new MQTT messages are received. Together, these components form an end-to-end data pipeline that turns OpenElectricity time-series data into a near real-time visual monitoring system for power generation and emissions.

2. Data Acquisition / Cleaning / Transformation

In this project, electricity generation and CO₂ emissions data were retrieved from the OpenElectricity API. The focus was on the National Electricity Market (NEM), and data was collected at 5-minute intervals for a selected period in May 2026 (e.g., from 1 May to 8 May 2026). The 5-minute granularity provides a high-resolution view of electricity production and emissions over time, which is essential for analysing temporal patterns and supporting real-time data streaming.

To address API rate limits and potential timeout issues, the data retrieval process was designed to operate in smaller time batches rather than a single large request. Specifically, the overall time range was divided into manageable intervals (e.g., hourly or daily requests), ensuring reliable responses and reducing the risk of request failures. This approach also improves scalability and robustness of the data acquisition pipeline.

The retrieved data was returned in a nested JSON structure, where each response contains multiple metrics (such as power and emissions), and each metric includes a list of time-series observations stored within a "results" field. As a result, a data transformation step was required to flatten this hierarchical structure into a tabular format suitable for downstream processing.

During preprocessing, several steps were performed. First, the nested "results" field was expanded so that each row represents a single observation at a specific timestamp. Second, timestamps were standardised into a consistent datetime format. Third, key attributes such as facility identifiers, metric types, and measurement values were extracted and reorganised into a structured schema. Any redundant or irrelevant fields were removed to simplify the dataset.

The cleaned and transformed data was then stored in CSV format. This materialisation step serves as a caching mechanism, allowing subsequent tasks (such as MQTT publishing and visualisation) to reuse the data without repeatedly calling the external API, thereby improving efficiency.

Additionally, because some power facilities consist of multiple operational units, the data was aggregated at the facility level. This aggregation ensures that each record represents the total power generation and emissions of a facility at a given timestamp. This step is particularly important for maintaining consistency when publishing data as individual MQTT messages, where each message is expected to correspond to a single facility at a specific point in time.

3. Data Integration

In assignment 1, we integrated the facility dataset with ABS dataset. Then, we create a map to show the location and attributes of facilities. During the processing, we understand displaying

a good visual representation in a map style requires the latitude and longitude of each facility to mark their location.

In assignment 2, we have some similar attributes such as power generation, fuel type and carbon dioxide emissions. Therefore, we keep these basic variables. They are partition by a narrower region area by network regions in NEM. Although we still need the region columns, each region's area shrinks to smaller area for more detailed analyses. Besides that, the time interval also changes from one year to five minutes. We need to change the time feature from year to datetime according to this change.

To implement the optional part, we should add additional features including the market price and market demand. These features are stored in new columns to represent them clearly. In the other hand, there are many feature columns that we used in assignment one is no longer appear at the dataset in assignment two. Therefore, we must find and remove these out-of-date features.

After keeping the remaining features columns, removing the out-of-date features, adding additional features, modifying the unit and datatype of region and time with changed scale and saving each facility's coordinates. We get the final assignment two scheme

Column	Pandas Dtype	Description
time	datetime64[ns]	Event time
facility_code	string	Facility id
facility_name	string	Facility name
network_region	category	Network region
fueltech_id	category	Fuel type
latitude	float64	Latitude
longitude	float64	Longitude
power_mw	float64	Power generation
emission_t	float64	CO2 emission
region_price_per_mwh	float64	Market price in region
region_demand_per_mwh	float64	Market demand in region

4. Data Visualisation

We designed to visualize the data by a dashboard map view. To achieve this, we need firstly download the background style of the map, the official network region boundary and, computed the initialized boundary of the map. We did some research and discovered open-source websites to support our project.

The regions of the map dashboard are drawn by the polygon from the downloaded geo-Json files. We also calculated the initial viewpoint on the map when initializing the map by the

regional polygons. The background of dashboard is the map image from the open-source provider.

Then, we mark all the facilities by their latitude and longitude. We use the dash library of python to add the click handler on each facility, the facility has no duplicate because we subscriber receive the data, it will keep the latest record with unique, facility code and fuel type.

When clicking the facility marker, there will show a window to display the facility's latest, power generation, emission and their corresponding fuel types. The window also shows the market price and market demand of the certain facility's region of the same event time.

Besides that, we add two manus for users to select the filter by region and fuel type. When choosing the specific filter option, the map will only show the facilities in corresponding groups of categories.

But use this dashboard, user can see the latest facility names, facility level power generation and carbon dioxide emissions and network region level market price and market demand with a adjustable region and fuel type.

5. Contribution

Yuting Xia (540242681): Worked with another group member on the code for Tasks 1, 2, 3 and 6, including retrieving data from OpenElectricity, combining the datasets, publishing MQTT messages, and setting up the continuous running process. I wrote the System Description section of the report.

Yiting Huang (540313349): Collaborated with a team member to complete Tasks 1, 2, 3, and 6, and contributed to the data acquisition section of the project report.

Jiayi Huang (540179745): Designed a schema suitable for assignment visualization. Write subscriber to receive data and store it according to the designed schema. Combining the received data with GeoJSON polygons from an open-source website, Visualized the map according to the assignment requirements.